MATH 102 - Applied Linear Algebra

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Contents

1 Invertible Matrix Theorem

1.1 Definition

Let A be a square n x n matrix. Then the following statements are equivalent. That is, for a given A, the statements are either all true or all false.

1. A is invertible.
2. A is row equivalent to I.
3. A has n pivot positions.
4. The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
5. The columns of A are linearly independent.
6. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.
7. The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$ in $\mathbb{R}^n$.
8. The columns of A span $\mathbb{R}^n$
9. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
10. There is an n x n matrix C such that CA = I.
11. There is an n x n matrix D such that AD = I.
12. A^T is invertible.
13. The columns of A form a basis of $\mathbb{R}^n$.
14. Col A = $\mathbb{R}^n$.
15. dim Col A = n.
16. rank A = n
17. Nul A = 0
18. dim Nul A = 0
19. The number 0 is *not* an eigenvalue of A.

20. det A $\neq 0$.

21. (Col A)$^\perp = \mathbf{0}$

22. (Nul A)$^\perp = \mathbb{R}^n$

23. Row A $= \mathbb{R}^n$

1.2 Thm 9.1.16

Let A $\in M_n$ and let $\lambda \in \mathbb{C}$. Then the following statements are equivalent:

(a) λ is an eigenvalue of A.

(b) A$\mathbf{x} = \lambda\mathbf{x}$ for some nonzero $\mathbf{x} \in \mathbb{C}^n$.

(c) (A - λI)$\mathbf{x} = \mathbf{0}$ has a nontrivial solution, that is, nullity(A - λI) > 0.

(d) rank(A - λI) $<$ n.

(e) A - λI is not invertible.

(f) A$^\top$ - λI is not invertible.

(g) λ is an eigenvalue of A$^\top$.

Proof

(a) $\Leftrightarrow$ (b)

By definition 9.1.1, if $A\mathbf{x} = \lambda\mathbf{x}$ and $\mathbf{x} \neq \mathbf{0}$ then $(\lambda, \mathbf{x})$ is an eigenpair of A, meaning λ is an eigenvalue of A.

(b) $\Leftrightarrow$ (c)

These are restatements of one another.

(c) $\Leftrightarrow$ (d)

By rank nullity theorem, if nullity(A - λI) > 0, then rank(A - λI) $<$ n.

(d) $\Leftrightarrow$ (e)

By property 1 and 16 of IMT, if rank(A - λI) $<$ n then (A - λI) is not invertible.

(e) $\Leftrightarrow$ (f)

Suppose we have $A - \lambda I = \begin{bmatrix} a-\lambda I & b \\ c & d-\lambda I \end{bmatrix}$, and suppose it is not invertible, its determinant is $(a-\lambda I)(d-\lambda I) - bc = 0$. Consider the case of $A^\top - \lambda I$, we have $\begin{bmatrix} a-\lambda I & c \\ b & d-\lambda I \end{bmatrix}$, whose determinant is $(a-\lambda I)(d-\lambda I) - cb$. And by commutativity of multiplication, $bc = cb$, so the determinant is also 0, meaning $A^\top - \lambda I$ is also not invertible.

(f) $\Leftrightarrow$ (g)

Since (a) is equivalent to (e), the same is true for $A^\top$.

2 Annihilators

2.1 Theorem 9.3.3

Let $A \in \mathbb{M}_n$ and let p be a nonconstant polynomial that annihilates A. Every eigenvalue of A is a root of $p(z) = 0$.

Proof:

Let $(\lambda, \mathbf{x})$ be an eigenpair of A. Then $\mathbf{0} = 0_n\mathbf{x} = p(A)\mathbf{x} = p(\lambda)\mathbf{x}$, so $p(\lambda)\mathbf{x} = 0$

2.1.1 Is The Converse True?

While all the eigenvalues of a matrix are roots of each of its annihilators, the roots of an annihilator are not necessarily eigenvalues.

This raises a question:

Must <u>some</u> of the roots of an annihilator of a matrix be eigenvalues of the matrix? The answer is yes.

Let $p(z) = a_0 + a_1 z + a_2 z^2 + \ldots + a_k z^k$ be an annihilator of a matrix $A \in \mathbb{M}_n$.
Factor p as follows: $p(z) = a_k(z-\lambda_1)(z-\lambda_2)\ldots(z-\lambda_k), a_k \neq 0$
Since p annihilates A, we have

$$0 = p(A) = a_k(A - \lambda_1 I)(A - \lambda_2 I)\ldots(A - \lambda_k I)$$

Since the zero matrix is not invertible, one of the factors $A - \lambda_i I$ is not invertible. This ensures that λ_i is an eigenvalue of A.

2.2 Each Matrix has an Eigenvector

2.2.1 Proof 1

λ is an eigenvector of $A \in \mathbb{M}_n$ iff $\det(A - \lambda I) = 0$.
$p(\lambda) = \det(A - \lambda I)$ is a polynomial of degree n (the characteristic polynomial of A).
Therefore, λ is an eigenvalue of A iff λ is a root of p (i.e. p)
By the Fundamental Theorem of Algebra (FTA), each polynomial over the complex has a root.
Hence, each matrix $A \in \mathbb{M}_n$ has an eigenvalue.

2.2.2 Proof 2

We've shown that each matrix $A \in \mathbb{M}_n$ has an annihilator p.
By the FTA, p must have a root.
In 2.1, we showed that at least one root of p is an eigenvalue of A.
Hence, each matrix $A \in \mathbb{M}_n$ has an eigenvalue.

3 Eigenvalues

3.1 Lemma 9.3.2

Let (λ, x) be an eigenpair of $A \in \mathbb{M}_n$ and let p be a polynomial. Then $p(A)x = p(\lambda)x$, that is, $(p(\lambda), x)$ is an eigenpair of $p(A)$.

Proof:
Recall $A^k\mathbf{x} = \lambda^k\mathbf{x}$.

$$\begin{aligned}
p(A)\mathbf{x} &= (c_kA^k + c_{k-1}A^{k-1} + ... + c_1A + c_0I)\mathbf{x} \\
&= c_kA^k\mathbf{x} + c_{k-1}A^{k-1}\mathbf{x} + ... + c_1A\mathbf{x} + c_0I\mathbf{x} \\
&= c_k\lambda^k\mathbf{x} + c_{k-1}\lambda^{k-1}\mathbf{x} + ... + c_1\lambda\mathbf{x} + c_0\mathbf{x} \\
&= (c_k\lambda^k + c_{k-1}\lambda^{k-1} + ... + c_1\lambda + c_0)\mathbf{x} \\
&= p(\lambda)\mathbf{x}
\end{aligned}$$

3.2 2-Commute

Let $A_1, A_2 \in \mathbb{M}_n$. Let λ be an eigenvalue of A_1. Suppose $A_1A_2 = A_2A_1$, then A_1, A_2 share an eigenvector in $\mathcal{E}_\lambda(A_1)$

Computation:

0. Choose A_1 or A_2. Say, A_1
1. Compute an eigenvalue of A_1. Say, λ, and form its eigenspace $\mathcal{E}_\lambda(A_1)$
2. Select a basis $\vec{x}_1, \vec{x}_2, ..., \vec{x}_k$ for $\mathcal{E}_\lambda(A_1)$ and form a matrix $X = [\vec{x}_1\ \vec{x}_2\ \ldots\ \vec{x}_k]$
3. Solve $A_2X = XC$ for C
4. Find an eigenpair of $C : (\mu, u)$
5. Compute $v = Xu$

Proof:

$$Ax_1 = \lambda x_1, \quad Ax_2 = \lambda x_2, \ ... \ , \ Ax_k = \lambda x_k$$
$$A\vec{x}_i = \lambda \vec{x}_i \qquad \text{for } i = 1, 2, ..., k$$

$$\begin{aligned} A_1 x_i &= \lambda x_i \\ A_2(A_1 x_i) &= A_2(\lambda x_i) \\ A_1(A_2 x_i) &= \lambda(A_2 x_i) && \text{(commutativity)} \end{aligned}$$

Case 1: $A_2 x_i = 0$
$\Rightarrow A_2 x_i \in \mathcal{E}_\lambda(A_1)$

Case 2: $A_2 x_i \neq 0$
$\Rightarrow A_2 x_i$ is an eigenvector of A_1 and $A_2 x_i \in \mathcal{E}_\lambda(A_1)$

$$A_2 x_1 = Xc_1, \ A_2 x_2 = Xc_2, \ ... \ , \ A_2 x_k = Xc_k$$
$$A_2[x_1 \ x_2 \ ... \ x_k] = [A_2 x_1 \ A_2 x_2 \ ... \ A_2 x_k] = [Xc_1 \ Xc_2 \ ... \ Xc_k] = X[c_1 \ c_2 \ ... \ c_k]$$
$$A_2 X = XC$$

$$\text{Let } (\mu, u) \text{ be an eigenpair of } C \qquad C\vec{u} = \mu\vec{u} \qquad u \neq 0$$
$$\text{Say } v = Xu \neq 0$$
$$Cu = \mu u$$
$$XCu = X\mu u$$
$$A_2 Xu = \mu Xu$$
$$A_2 v = \mu v$$
$$\therefore v = Xu \in \mathcal{E}_\lambda(A_1) \text{ is an eigenvector of } A_2$$

3.3 Gershgorin Disk Theorem

If $n \geq 2$ and $A = \left[a_{ij}\right] \in \mathbb{M}_n$ then

$$\text{spec } A \subseteq \mathcal{G}(A) = \bigcup_{k=1}^{n} \mathcal{G}_k(A).$$

Proof:

Let $(\lambda, \mathbf{x})$ be an eigenpair of $A = \left[a_{ij}\right] \in \mathbb{M}_n$ and let $\mathbf{x} = \left[x_i\right]$. Since $\lambda\mathbf{x} = A\mathbf{x}$, we have

$$\lambda x_i = \sum_{j=1}^{n} a_{ij}x_j = a_{ii}x_i + \sum_{j \neq i} a_{ij}x_j$$
$$(\lambda - a_{ii})x_i = \sum_{j \neq i} a_{ij}x_j$$

Let $k \in \{1, 2, ..., n\}$ be any index such that $|x_k| = \max_{1 \leq i \leq n} |x_i|$. Since $\mathbf{x} \neq \mathbf{0}$, it follows that $|x_k| > 0$ and $\frac{|x_j|}{|x_k|} \leq 1$ for all $j = 1, 2, ..., n$. Set $i = k$ and divide by x_k to obtain

$$\lambda - a_{kk} = \sum_{j \neq k} a_{kj}\frac{x_j}{x_k}$$

By the triangle inequality,

$$|\lambda - a_{kk}| = \left|\sum_{j \neq k} a_{kj}\frac{x_j}{x_k}\right| \leq \sum_{j \neq k} |a_{kj}| \left|\frac{x_j}{x_k}\right| \leq \sum_{j \neq k} |a_{kj}| = R_k'(A)$$

Therefore, $\lambda \in \mathcal{G}_k(A) \subseteq \mathcal{G}(A)$

3.4 Corollary 9.4.15

Let $A \in \mathbb{M}_n$. If A is strictly diagonally dominant, then it is invertible.

Proof:

The strict diagonal dominance of A ensures that $0 \notin \mathcal{G}_k(A)$ for each $k = 1, 2, ..., n$. The Gershgorin Disk Theorem then tells us that 0 is not an eigenvalue of A and therefore A is invertible.

4 Similar Matrices

4.1 Theorem 9.6.3

If two real matrices are similar, then they are similar via a real matrix.

Proof:

Let $A, B \in \mathbb{M}_n(\mathbb{R})$. Let $S \in \mathbb{M}_n$ be invertible such that $A = SBS^{-1}$. For any $\theta \in (-\pi, \pi]$, let $S_\theta = e^{i\theta}S$ and compute $A = (e^{i\theta}S)B(e^{-i\theta}S^{-1}) = S_\theta B S_\theta^{-1}$. Then $AS_\theta = S_\theta B$. The complex conjugate of this identity is $A\bar{S}_\theta = \bar{S}_\theta B$ since A and B are real. Let $R_\theta = S_\theta + \bar{S}_\theta = 2\ \mathrm{Re}\ S_\theta$ which is real for all $\theta \in (-\pi, \pi]$. Then $AR_\theta = A(S_\theta + \bar{S}_\theta) = (S_\theta + \bar{S}_\theta)B = R_\theta B$. The computation

$$R_\theta = S_\theta(I + S_\theta^{-1}\bar{S}_\theta) = S_\theta(I + e^{-2i\theta}S^{-1}\bar{S}) = e^{-2i\theta}S_\theta(e^{2i\theta}I + S^{-1}\bar{S})$$

shows that R_θ is invertible if $-e^{-2i\theta}$ is not an eigenvalue of $S^{-1}\bar{S}$. Since $S^{-1}\bar{S}$ has at most n distinct eigenvalues, there is a ϕ such that $-e^{-2i\phi}$ is not an eigenvalue of $S^{-1}\bar{S}$. Then $AR_\phi = R_\phi B$, where R_ϕ is real and invertible, and $A = R_\phi B R_\phi^{-1}$.